AI23521 BUILD AND DEPLOY FOR MACHINE LEARNING APPLICATION

PROJECT: STUDENT PASS/FAIL PREDICTION USING MACHINE LEARNING

AIM:

To develop a machine learning application that predicts whether a student will pass or fail using study hours, attendance, and previous exam score with feature scaling and a Logistic Regression classifier.

ABSTRACT:

Early identification of students at risk of failing helps teachers provide timely support. This project demonstrates a binary classification pipeline that learns from labelled student records and predicts the result of a new student. Numerical features are standardized using StandardScaler and classified using Logistic Regression. Python and scikit-learn are used. A larger real student dataset can be used to extend the demonstration.

PROPOSED SYSTEM:

The system accepts a student's study hours, attendance percentage and previous exam score, scales the features, and passes them to a trained classifier. The predicted label is displayed as PASS or FAIL. A small built-in example dataset allows the program to run without downloading external data. Since this dataset is intentionally small, its accuracy is not representative of real-world performance.

ALGORITHM:

1. Import the required Python libraries.
2. Prepare labelled student records belonging to pass and fail classes.
3. Split the records and labels into training and testing sets.
4. Standardize the numerical features using StandardScaler.
5. Train a Logistic Regression classifier on the training features.
6. Predict test labels and calculate accuracy.
7. Transform a new student record using the same scaler and predict the result.
8. Display the predicted result.

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CODE:

```
# Student Pass/Fail Prediction using Logistic Regression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score

# Each row: [study hours per day, attendance %, previous exam score]
X = [
    [1, 55, 40], [2, 60, 45], [1, 50, 35], [2, 65, 50],
    [3, 58, 42], [1, 62, 48], [2, 52, 38], [3, 64, 46],
    [2, 57, 41], [1, 59, 44],
    [5, 85, 75], [6, 90, 80], [4, 80, 70], [7, 92, 85],
    [5, 88, 78], [6, 82, 72], [4, 86, 68], [7, 95, 90],
    [5, 78, 74], [6, 84, 79]
]

y = ["fail"] * 10 + ["pass"] * 10

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)

model = Pipeline([
    ("scaler", StandardScaler()),
    ("classifier", LogisticRegression())
])

model.fit(X_train, y_train)

predictions = model.predict(X_test)
print("Test accuracy:", round(accuracy_score(y_test, predictions), 2))

new_student = [[5, 82, 70]]
prediction = model.predict(new_student)[0]
print("Student [hours, attendance, previous score]:", new_student[0])
print("Predicted result:", prediction.upper())
```

SAMPLE OUTPUT:

```
Test accuracy: 1.0
Student [hours, attendance, previous score]: [5, 82, 70]
Predicted result: PASS
```

Note: The test accuracy depends on the selected split, dataset and scikit-learn version. The small example dataset is for demonstrating the workflow only.

RESULT:

The Student Pass/Fail Prediction application was implemented using feature scaling and a Logistic Regression classifier. The system trains on labelled student records and predicts whether a new student is likely to pass or fail. Thus, a basic end-to-end machine learning workflow was demonstrated using Python.